

CANDIDATE NAME: _____

INDEX NUMBER _____



SERANGOON JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2017

BIOLOGY 9744
Higher 2

CG _____

PAPER 1

Thursday
21 September 2017

1 hour

Additional materials:
Optical Mark Sheet

INSTRUCTIONS TO CANDIDATES

Write your name and CG in the spaces at the top of this page.

On the Optimal Mark Sheet, enter your name, subject title, test name, class. For your index number, enter your full NRIC number. Shade the corresponding lozenges on the OMS according to the instructions given by the invigilators.

AT THE END OF THE EXAMINATION, HAND IN BOTH THE OMS AND QUESTION PAPER.

INFORMATION FOR CANDIDATES

There are **thirty (30) questions** in this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C, D**. Choose the **one** you consider correct and record your choice in **soft pencil** on the OMS.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done on the question paper.

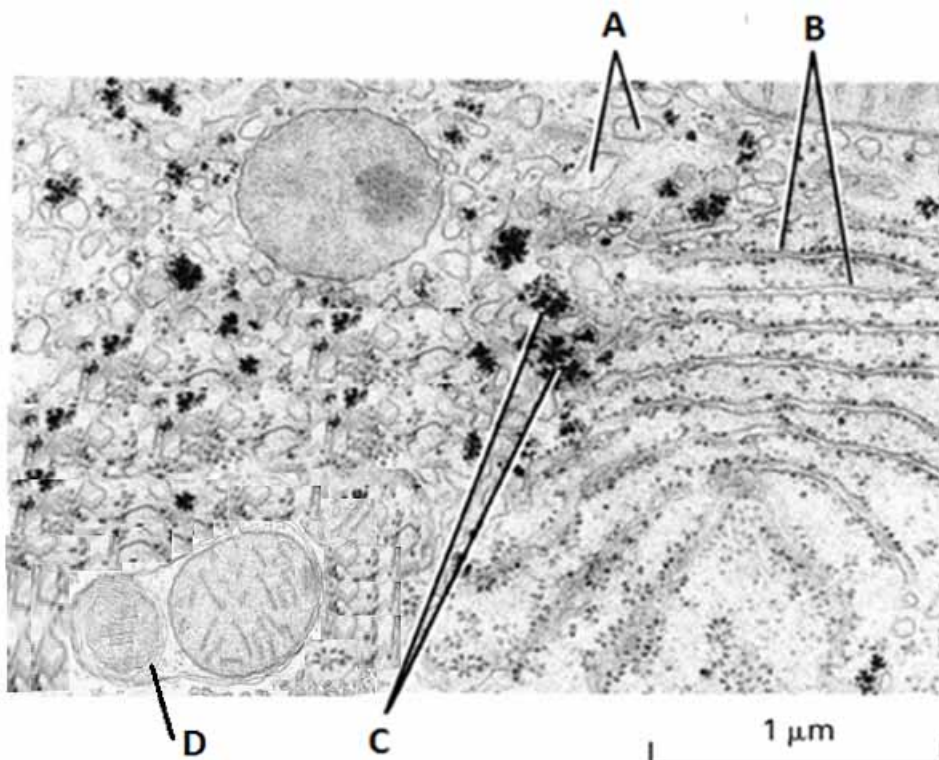
This question paper consists of 16 printed pages.

Answer all questions on the OTAS provided.

- 1 EDTA is used extensively as an anticoagulant for stored blood in blood banks. Thrombokinase plays a major role in the clotting of blood. EDTA decreases the reaction rate of thrombokinase by binding to calcium ions.

Which of the following describes the role of calcium ions?

- A Allosteric inhibitors
 - B Coenzymes
 - C Cofactors
 - D Competitive inhibitors
- 2 A cell in the G1 phase has two homologous pairs of chromosomes. It then undergoes two mitotic divisions. At the end of the second mitotic division, what is the total number of chromosomes and gene loci found in all the daughter cells formed?
- A 8 chromosomes and 4 times as many gene loci as the original parent cell.
 - B 8 chromosomes and 8 times as many gene loci as the original parent cell.
 - C 16 chromosomes and 4 times as many gene loci as the original parent cell.
 - D 16 chromosomes and 8 times as many gene loci as the original parent cell.
- 3 The electron micrograph below shows a liver cell.

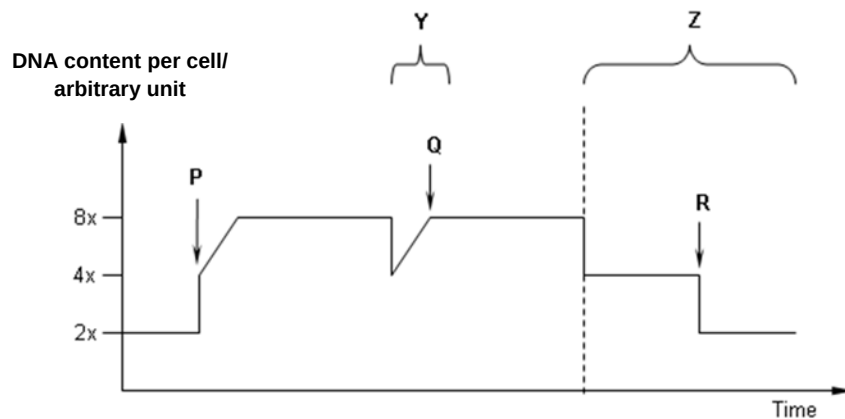


Which statement(s) correctly describe(s) the labelled structures?

- 1 Structure **A** transports proteins from Structure **B** to Golgi Apparatus.
- 2 Proteins enter the lumen of Structure **B**, where they undergo chemical modifications such as glycosylation.
- 3 Structure **C** is starch grain.
- 4 The process shown in structure **D** is autolysis.

A 2 only **B** 1 and 2 only **C** 2 and 3 only **D** 2, 3 and 4 only

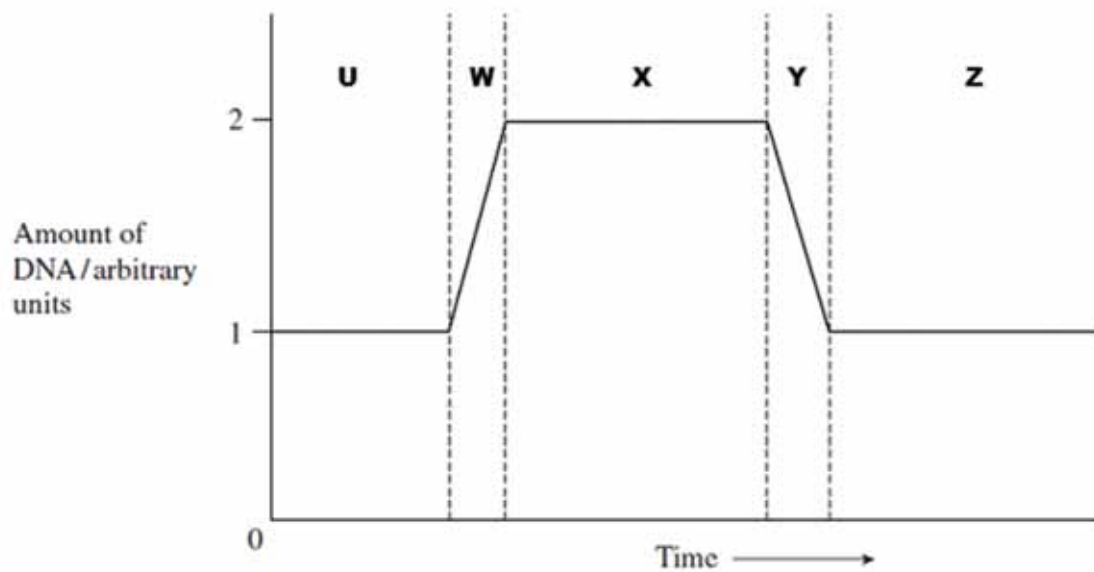
- 4 The graph represents the changes in the DNA content within a cell at different stages in the cell cycle.



Name the events occurring at **P**, **Q** and **R**, and identify the stage where meiosis is occurring.

	P	Q	R	Meiosis occurring at
A	S phase	Fertilisation	Cytokinesis	Y
B	Fertilisation	Interphase	Cytokinesis	Z
C	S phase	Prophase	Telophase	Y
D	Fertilisation	Metaphase	Telophase	Z

- 5 The graph shows changes in the amount of DNA in a cell during one cell cycle. The letters U – Z marks out the different phases in the cell cycle.



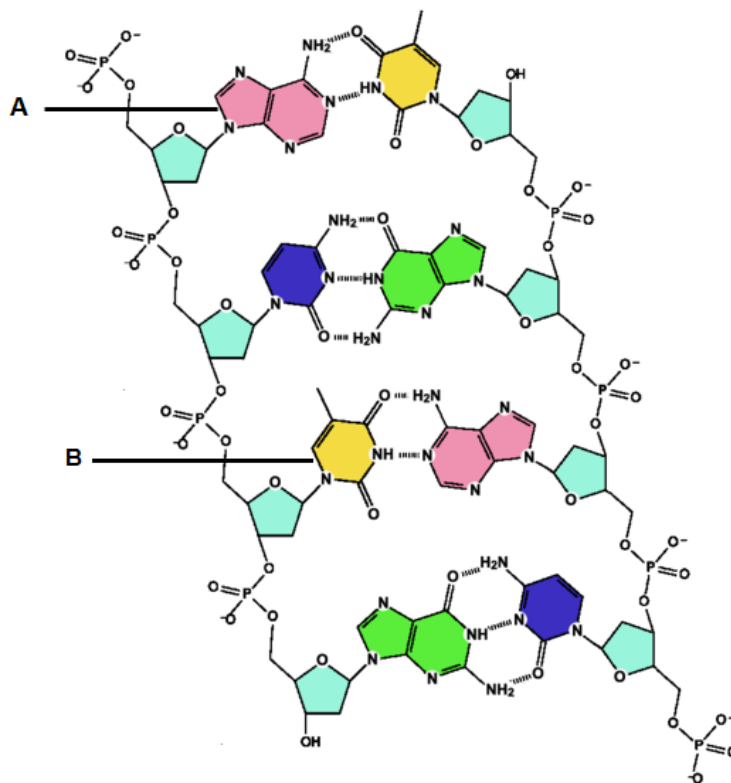
Many drugs that are used to treat cancer work at different time periods during the cell cycle.

- (i) Cisplatin binds to DNA and stops free DNA nucleotides from joining together.
- (ii) Drug B stops spindle fibres from shortening.

With reference to the cell cycle above, determine where these 2 drugs work.

	Cisplatin	Drug B
A	W	X
B	W	Y
C	U	X
D	U	Z

6 The figure below shows a DNA molecule.



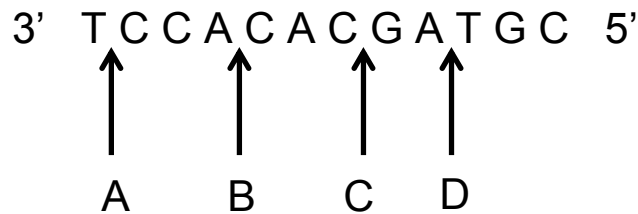
Which statement(s) correctly describe the polynucleotide?

- 1 The structure labelled **A** corresponds to that of a purine, while the structure labelled **B** corresponds to that of a pyrimidine.
- 2 The antiparallel nature of DNA double helix allows phosphodiester bonds to form between the nitrogenous bases of opposite strands.
- 3 Distance between adjacent deoxyribonucleotides is 3.4 Å and one turn consists of 10 deoxyribonucleotides. (Note: 10 Å = 1 nm)
- 4 The wound DNA double helix consists of alternating major grooves and minor grooves along its axis which are essential for the binding with proteins.

- A** 1 only
B 1 and 2 only
C 3 and 4 only
D 1, 3 and 4 only

- 7 The RNA triplet UAG acts as a stop codon terminating the synthesis of a polypeptide. The diagram shows a template strand of DNA which codes for four amino acids.

Where would a mutation, introducing a thymine nucleotide, result in the termination of translation?



- 8 Which of the following is **not** a feature of eukaryotic gene expression?

- A Polycistronic mRNAs are very rare.
- B Many genes are interrupted by noncoding DNA sequences.
- C RNA synthesis and protein synthesis are coupled.
- D mRNA is often extensively modified before translation.

- 9 Human telomeres consist of repeating TTAGGG sequences which extend from the ends of the chromosomal DNA. When cells undergo mitotic division, some of these repeating sequences are lost. This results in a shortening of the telomeric DNA.

What is a consequence of the loss of repeating DNA sequences from the telomeres?

- A The cell will begin the synthesis of different proteins.
- B The cell will begin to differentiate as a result of the altered DNA.
- C The number of mitotic divisions the cell can make will be limited.
- D The production of mRNA will be reduced.

- 10** The translation mixture contains a polynucleotide that directs the synthesis of Met-Gly-Gly-Phe-Leu-Ala. In the presence of Azithromycin, this polymer directs the synthesis of Met-Gly only.

From the information given, which of the following deductions could you make about Azithromycin?

	Control Stage	Conclusion
A	Translational	It prevents formation of the initiation complex, which contains the initiator tRNA and both ribosomal subunits.
B	Post translational	It inhibits binding of aminoacyl- tRNAs to the A site in the ribosome.
C	Translational	It blocks translocation of peptidyl transferase-rRNA from the A site to the P site of the ribosome.
D	Post translational	It interferes with chain termination and release of the peptide.

- 11** Which of the following statement(s) about cancer is / are true?

- I Individuals who inherit one mutant tumour suppressor gene are more likely to develop cancer than individuals with two non-mutant copies.
- II Cancer is a result of increased cell division which promotes the mutation of a proto-oncogene.
- III Mutagenic activation of a single oncogene is sufficient to cause a normal cell to develop into a cancerous cell.

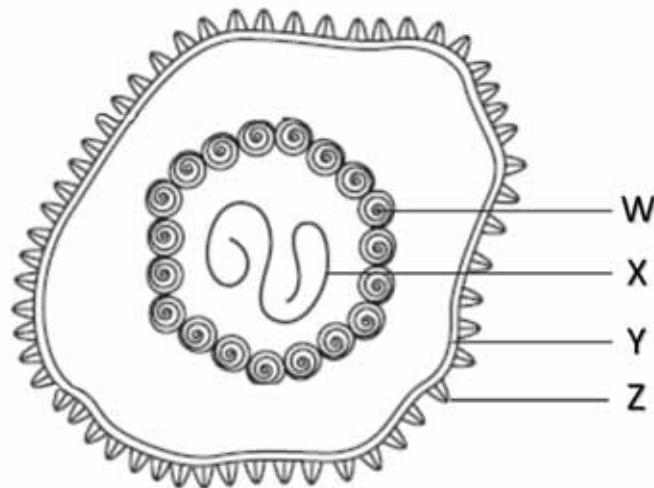
- A** I only
- B** I and II only
- C** I and III only
- D** I, II and III

- 12** To date, more than 10 different strains of influenza virus (e.g. H1N1, H2N3, H5N1, H7N9 and so on) have been documented.

Which of the following structural characteristic of influenza virus makes this possible?

- A** Single-stranded RNA as its genetic material
- B** Presence of an envelope that is derived from the host cell
- C** Eight separate segments of genetic material
- D** Presence of error-prone reverse transcriptase within the virus

13 The figure below shows the structure of a virus.



Which of the following matches the functions of structures W – Z?

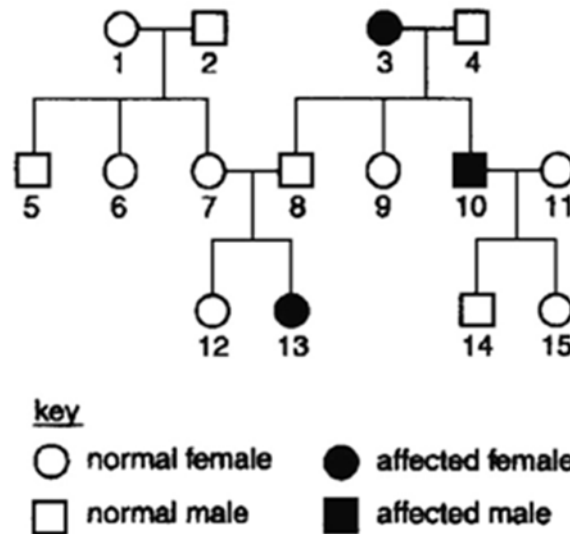
	W	X	Y	Z
A	Ensures the integrity of the viral genome is maintained	Entry of virus into host cell	Specificity of host cell	Assembly of viruses
B	Ensures the integrity of the viral genome is maintained	Assembly of viruses	Entry of virus into host cell	Specificity of host cell
C	Specificity of host cell	Assembly of viruses	Ensures the integrity of the viral genome is maintained.	Entry of virus into host cell
D	Assembly of viruses	Ensures the integrity of the viral genome is maintained.	Entry of virus into host cell	Specificity of host cell

14 When the *lac* operon for lactose metabolism is switched off, which of the following genes would still be expressed?

- I β -galactosidase gene
- II RNA polymerase gene
- III CAP gene
- IV Repressor gene

- A I and II
- B I and III
- C II, III and IV
- D All of the above

- 15 The pedigree chart below shows the inheritance of a recessive condition known as human albinism. Only homozygous recessive individuals are albinos.



What is the probability of individual 9 being a heterozygous carrier?

- A 0.00
 - B 0.25
 - C 0.50
 - D 1.00
- 16 Which of the following regarding embryonic stem cells and hematopoietic stem cells is true?
- A As embryonic stem cells develop, they turned into hematopoietic stem cells as they lose their ability to differentiate into all types of cells.
 - B Embryonic stem cells have more genes than hematopoietic stem cells and thus are able to form more types of cells.
 - C Under normal conditions, embryonic stem cells express more of their genes compared to the hematopoietic stem cells.
 - D Both stem cells are derived from the zygotic stem cells with the hematopoietic stem cells having a lowered telomerase activity compared to the embryonic stem cells.

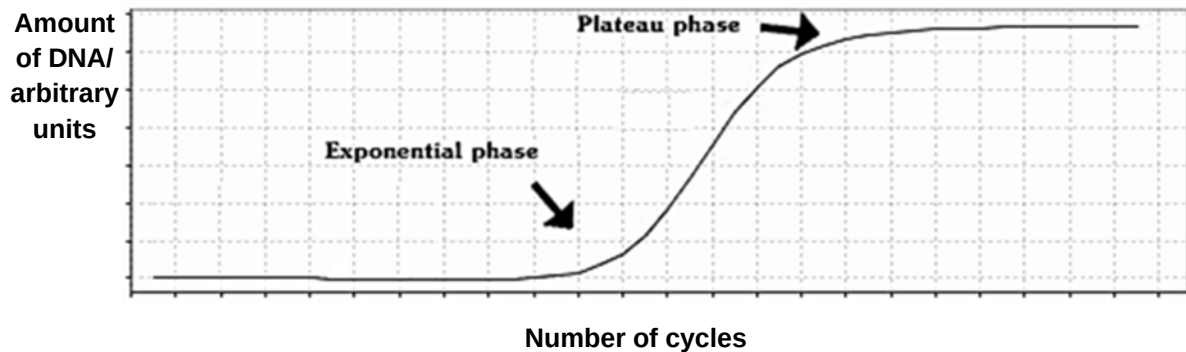
- 17** A plant researcher tried to investigate a cross between two heterozygous Snapdragon plants that produced red flowers. She predicted three possible phenotypic outcomes, namely plants with white flowers, pink flowers and red flowers, with a phenotypic ratio of 4:3:9 respectively. When the cross was performed, she found 50 plants with white flowers only, 41 plants with pink flowers, and 85 plants with red flowers. A chi-squared test was performed, and the chi-squared value was calculated to be 4.74

Degree of freedom	Probability, P				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

Which of the following statements is correct?

- A** The degree of freedom is 3.
 - B** The calculated chi-squared value is greater than the critical chi-squared value.
 - C** There is a high probability that the difference between the observed and expected values is due to chance.
 - D** The probability that the difference between observed and expected values is due to chance is less than 5%.
- 18** Recent advances in the field of stem cell research have shown that induced pluripotent stem cells (iPS cells) can be artificially derived from adult somatic cells. iPS cells are mostly similar to natural pluripotent cells. This implies that iPS cells can
- A** theoretically differentiate into all cell types.
 - B** theoretically differentiate into any of the three germ layers.
 - C** theoretically differentiate into gametes.
 - D** theoretically capable of transdifferentiation.

- 19 During the process of polymerase chain reaction (PCR), the amount of DNA synthesised can be traced using fluorescent probes and the measurements are shown in the following plot. The process initially goes through an exponential phase followed by a plateau phase eventually.



Which of the following statements is **true**?

- A During the exponential phase, the number of DNA molecules synthesized after 15 cycles is 15^2 .
 - B During the exponential phase, the temperature is always maintained at the optimum temperature of 72°C hence there is rapid amplification.
 - C During the plateau phase, the reaction mixture is being depleted of ribonucleotides.
 - D During the plateau phase, *Taq* polymerase may be denatured.
- 20 The dashed lines in the template sequence represent a long sequence of bases to be amplified.

Template

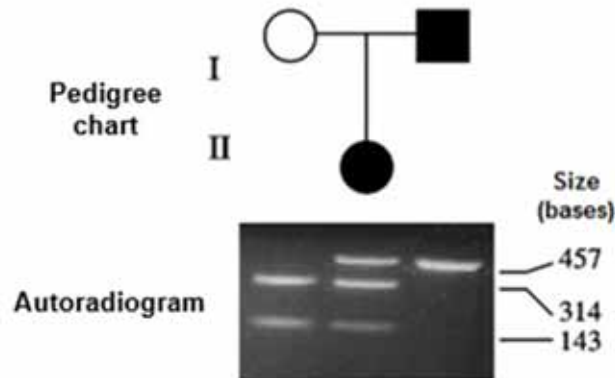
5' ATTCGGACTTG ----- GTCCAGCTAGAGG 3'

3' TAAGCCTGAAC ----- CAGGTCGATCTCC 5'

Which of the following sets of primers can be used in the PCR for the amplification of the following DNA sequence?

- A 5' GTCCAGC 3' & 5' CCTGAAC 3'
- B 5' ATTCGGA 3' & 5' CCTCTAG 3'
- C 5' GGAAGTTG 3' & 5' GCTGGAC 3'
- D 5' AUUCGGA 3' & 5' GAUCUCC 3'

- 21** A family with a history of a genetic disease is studied using restriction digestion of the DNA samples containing the gene responsible for the disease. The pedigree chart of the family is aligned with the autoradiogram obtained from Southern blotting. (Shaded symbols in the pedigree chart indicate individuals affected by disease.)



Based on the information given, which of the following can be deduced?

- A** The disease allele is dominant to the normal allele.
 - B** The mutation creates a new restriction site in the affected gene.
 - C** One of the parents in generation I is a carrier.
 - D** The offspring in generation II is a carrier.
- 22** Which of the following statements correctly compares oxidative phosphorylation and non-cyclic photophosphorylation?
- A** Both types of phosphorylation produce ATP and oxygen as end products.
 - B** Both types of phosphorylation produce ATP and the reduced form of a redox reagent.
 - C** Oxidative phosphorylation is involved in the conversion of one form of chemical energy to another while non-cyclic photophosphorylation is involved in converting light energy to chemical energy.
 - D** Water is an electron donor in non-cyclic photophosphorylation while it is an electron acceptor in oxidative phosphorylation.
- 23** What happens to most of the reduced NAD molecules in cell metabolism?
- A** They act as oxidising agents in glycolysis.
 - B** They are oxidised in inner mitochondrial membrane for ATP formation.
 - C** They are oxidised in the Calvin cycle.
 - D** They combine with succinic acid as part of Krebs cycle.

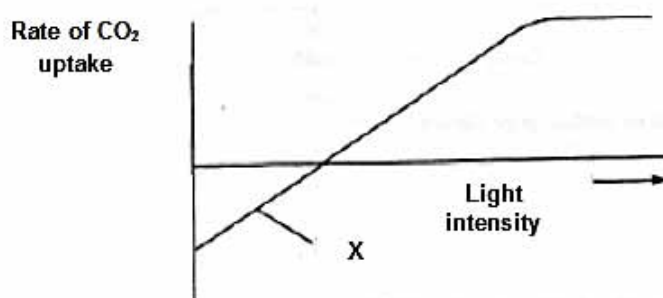
- 24 Rotene and oligomycin are two metabolic poisons which affect cellular respiration. The effects of rotene and oligomycin on aerobic respiration are summarised in the table.

	Ability to use glucose	Ability to use oxygen	ATP yield
Rotene	Yes	No	Decreases
Oligomycin	Yes	Yes	Decreases

Which of the following correctly identifies the specific functions of these two metabolic poisons?

	Rotene	Oligomycin
A	Electron transport inhibitor	Inhibits ATP synthase
B	Inhibits ATP synthase	Electron transport inhibitor
C	Dissipate proton gradient	Inhibits ATP synthase
D	Inhibits ATP synthase	Dissipate proton gradient

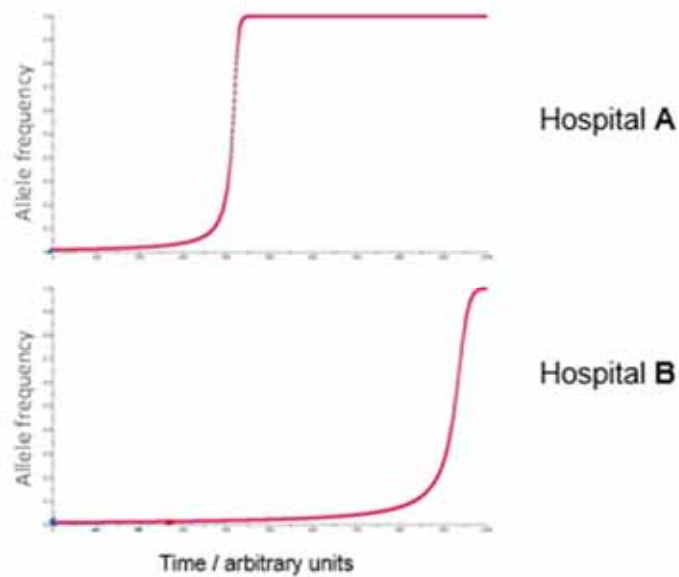
- 25 In the graph below, the rate of CO₂ uptake by plant cells is shown to vary with increasing light intensity.



Which of the following is true at point X?

- A** The plant is photosynthesizing.
- B** Rate of respiration equals rate of photosynthesis.
- C** CO₂ is a limiting factor.
- D** There is not enough light for photosynthesis to have commenced.

- 26 The two graphs below show the allele frequency of an antibiotic resistance gene *Neo* in the gene pool of *Streptococcus pneumoniae*, a bacteria that causes pneumonia.



Which of the following statements can be concluded from the graphs?

- A There is more genetic variation in the gene pool of *Streptococcus pneumoniae* in hospital **A** than hospital **B**.
 - B Patients in hospital **A** were treated with antibiotic Neomycin more frequently than patients in hospital **B**.
 - C The rate of mutation in the genome of *Streptococcus pneumoniae* in hospital **B** occurs more slowly than that in hospital **A**.
 - D Patients in hospital **A** has a stronger immune system than patients in hospital **B**.
- 27 Which sequence of events correctly describes evolution?

- 1 Differential reproduction of the spiders occurs.
- 2 A new selection pressure occurs.
- 3 Allele frequencies within the spider population change.
- 4 Poorly adapted spiders have decreased survivorship.

- A 2 4 1 3
- B 2 4 3 1
- C 4 1 3 2
- D 4 3 1 2

28 The following statements are some findings of scientists in an attempt to investigate the evolutionary relationship between the anteater, armadillo and pangolin.

- I** Anteater, armadillo and pangolin feed primarily on insects such as ants.
- II** Anteater, armadillo and pangolin have long tongue and strong digging limbs.
- III** The tongues of the anteater and armadillo are connected to the hyoid bone while the tongue of pangolin is not.
- IV** There is a higher percentage similarity between the DNA sequences of Anteater and armadillo than with the pangolin.
- V** There is very low percentage similarity between the DNA sequences of anteater and pangolin as well as between the armadillo and pangolin.

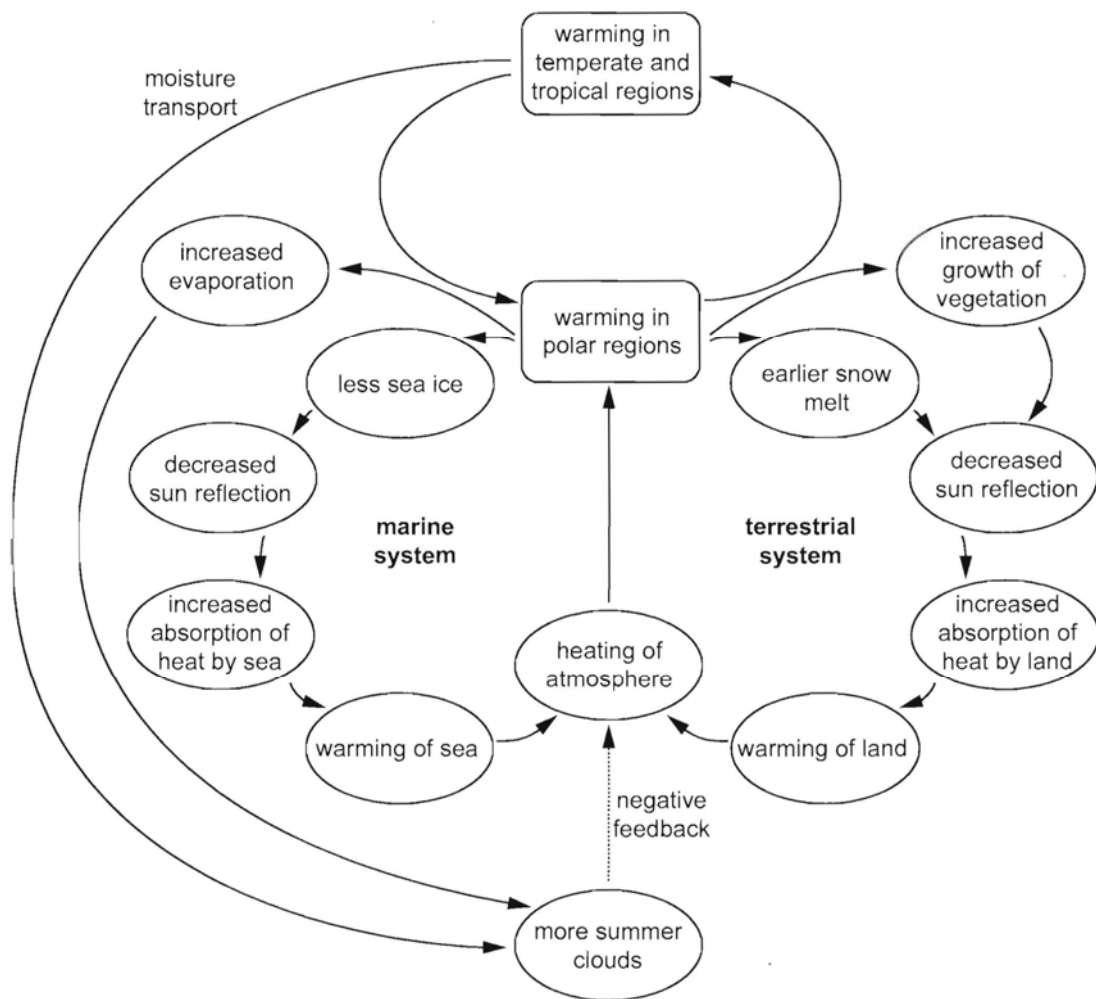
Which of the following conclusions can be drawn from the statements given above?

- A** The anteater and pangolin have experienced divergent evolution as shown by homologous structures between their hyoid bones and tongues.
- B** The anteater and pangolin have experienced convergent evolution as shown by homologous structures in their hyoid bones and tongues.
- C** The armadillo and pangolin have experienced divergent evolution as shown by the low similarity between their DNA sequences.
- D** The anteater and armadillo have experienced divergent evolution as shown by similarities in their DNA sequences and homologous anatomical structures.

29 In order to initiate an adaptive immune response, antigenic peptide must be presented to antigen-specific T cells. Which one type of cell presents this antigen to T cells?

- A** Dendritic cell
- B** Epithelial cell
- C** Neutrophil
- D** Plasma cell

- 30 The diagram shows the effect of increasing temperatures on the ice and snow cover at the polar regions.



Which effect of higher temperatures in the polar regions could increase global warming?

- A Increased evaporation leads to more rainfall, which absorbs heat from the land and sea.
- B Melting of ice and snow results in less reflection of sunlight and more heat absorption by the Earth.
- C Melting of sea ice causes more cloud formation, which increases absorption of heat in the atmosphere.
- D Earlier melting of snow allows vegetation cover to increase faster, reducing loss of heat from the surface of the Earth.

End of Paper